

Specialization:

Logistics

Business Focus:

Logistics industry

Skill Outcome:

- API integration

SkyRide Logistics Ltd

Leveraging Weather APIs for Smarter Business Decisions

A Data Engineering CASE STUDY

Project Learning Opportunities:

Introducing the student to how api is a core aspect of data engineering showing them how it can be integrated into a web app

Tools and Technology Used:

Company Introduction

SkyRide Logistics Ltd. is a mid-sized logistics company that provides same-day and intercity delivery services across major Nigerian cities. The company operates a fleet of delivery vans and motorcycles, serving both corporate clients and individual customers.

Core Operations Include:

- Daily parcel and document deliveries
- E-commerce order fulfillment for partner merchants
- Time-sensitive logistics (medical, food, and retail deliveries)
- Route planning and vehicle dispatching



The Business Problem

SkyRide is experiencing deliveries delays and sudden weather changes

Your Role

You have been engaged as a Software Engineer to support SkyRide. Your responsibility is to implement a system for weather changes by integrating an API to get current weather updates.

The Problem

Purpose: Break the problem into measurable components.

- The company's drivers often encountered sudden heavy rainfall and poor road visibility.
- Deliveries were delayed, resulting in customer dissatisfaction and financial loss.
- Operations team relied on manual weather checks, which were inconsistent and not location-specific.
- The company needed a real-time, automated weather update system to optimize delivery routes and scheduling.



Rationale for the Project

This Project should....

Rationale 1



Operational Efficiency and Risk Reduction

Unpredictable weather conditions were causing frequent delivery delays and damaged goods.

Rationale 2



Customer Experience and Retention
Late deliveries and poor communication negatively affected customer trust. By leveraging real-time weather data, the company can send timely notifications, set accurate delivery expectations, and maintain transparency — all of which improve satisfaction and loyalty.

Rationale 3



Before the project, route planning was largely manual and reactive. With a Weather API, the company gains access to continuous, location-specific weather insights, enabling smarter planning, optimized fuel usage, and better resource allocation across operations.

Project Workflow

Step 1

Set up Api configuration(API key)

Step 2

Understand how to go through documentation

Step 3

Set up Api configuration to make the calls

Step 4

Show how this requests can be integrated into a web app

Let's Dive in

Please note

When you are done with this case study:

Kindly **share your work on GitHub**, and don't forget to **add it to your portfolio** or **share on LinkedIn**